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An affective computing framework for preschool music education using MoE architecture and neural radiance fields

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Abstract

The key content of this paper is the digital construction of preschool music education scenarios and affective computing. It adopts an innovative approach, integrating the MoE architecture and neural radiance field technology together. The Edu-NeRF reconstruction framework is relied on to achieve high-fidelity reconstruction of dynamic scenes, and its PSNR value is 28.5. The SSIM value is 0.92, which is 10.8% higher than that of the traditional method. This paper also constructed an affective computing model. After training this model with 60 h of multimodal data from 120 preschool children, its accuracy rate for emotional state recognition reached 89.7%, with a macroscopic F1 value of 0.881. By combining cross-modal fusion algorithms, we have successfully broken through the technical bottlenecks in dynamic scene reconstruction and affective computing. In terms of real-world applicability, this approach significantly outperforms conventional 2D music education products. It enables real-time, high-fidelity interaction on standard low-power devices, thereby reducing hardware deployment costs and lowering the barrier for widespread school adoption. Furthermore, experimental verification demonstrates that this model increases children's learning engagement and the efficiency of personalized interaction by 30%. By enhancing both the immersive learning experience and teacher acceptance, this framework provides a highly practical and scalable technical path for the digital transformation of early childhood education.

Keywords MoE architecture, Neural radiance fields, Preschool music education, Dynamic scene reconstruction

1 Introduction

With the in-depth implementation of the “Digital Education” strategy, the integration of technology and education is constantly deepening. Preschool music education, as a key part of aesthetic education enlightenment, has long relied on teachers’ personal experience and intuition in the construction of teaching scenarios and in the perception of students’ learning states, lacking objective, quantitative evaluation methods. Traditional online music education products primarily feature 2D videos or simple animation



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interactions, which suffer from key problems, including limited immersion, a single interaction dimension, and an inability to capture and respond to the complex emotional changes of preschool children. Specifically, these existing products fail to account for the unique psychological and behavioral challenges of this age group. Preschoolers typically have short attention spans, exhibit high expressive variability in their uncoordinated body language, and face significant difficulties in accurately verbalizing their inner feelings. Consequently, relying on a single interaction dimension or traditional unimodal evaluation is entirely inadequate for this demographic, which strongly justifies the urgent need for a multimodal affective computing approach specifically tailored to preschoolers.

In recent years, breakthrough advances in Neural Radiance Fields (NeRF) technology have established a new paradigm for reconstructing three-dimensional scenes with high realism and free-viewpoint browsing [1]. By learning a continuous scene representation, it can synthesise realistic images from sparse two-dimensional image inputs from any perspective, laying a technical foundation for the construction of highly immersive virtual education environments. However, the standard NeRF model is primarily designed for static scenes, and its reconstruction performance in educational scenarios with dynamic characters is not ideal.

On the other hand, affective computing is the key to achieving personalized education. Preschool children's emotional states are manifested in multiple modalities: facial expressions and body movements, humming and rhythm, as well as heart rate and galvanic skin response [2]. Fusing heterogeneous multimodal information and accurately identifying the emotional dimensions of the learning state are significant challenges. The Mixture of Experts (MoE) model, as a sparsely activated architecture, can process different data patterns or subtasks via multiple "expert" networks, with the gating network determining the weight distribution, demonstrating great potential for handling complex, multi-source data [3].

2 Theoretical basis and related research

2.1 Neural radiance fields and their variants

NeRF represents a continuous scene as a multi-layer perceptron. It takes a three-dimensional spatial position and a two-dimensional viewing direction as inputs and outputs volume density and colour. New-perspective images are synthesised through volume rendering technology [5, 6]. Subsequent studies have improved it to address issues such as slow training speed and the inability to handle dynamic scenarios. D-NeRF introduces a time variable to handle dynamic scenarios. NR-NeRF decomposes the dynamic scene into a canonical scene representation and a deformation field, improving the ability to model non-rigid motion [7]. The Edu-NeRF in this paper draws on such ideas but is specifically optimised for the characteristics of a "relatively static background" and "specifically dynamic preschool children" in educational scenarios [8].

2.2 Application of mixture of experts model in multimodal learning

The MoE model processes each input sample by activating a subset of its parameters, avoiding a quadratic increase in computational complexity while expanding the model's capacity [9]. The Switch Transformer successfully applies the MoE idea to the Transformer architecture, demonstrating its advantages in ultra-large-scale models [10]. In

the field of multimodal learning, MoE enables different experts to handle distinct modalities or feature levels [11]. However, existing work mainly focuses on adult emotional recognition or general multimodal tasks, and the design of MoE models for preschool children in music education scenarios remains insufficiently explored.

2.3 Emotional computing in education scenarios

Educational affective computing aims to identify students' cognitive-affective states, such as confusion, engagement, and boredom [12]. Traditional methods are primarily based on a single modality, such as using facial expression analysis or classroom behaviour analysis. In recent years, multimodal fusion has become a trend. For example, visual and audio information are combined to judge students' classroom participation [13]. However, these existing methods exhibit significant limitations when directly applied to preschool scenarios. First, traditional vision-based affective computing struggles with the frequent, non-rigid motions typical of active young children, often leading to tracking failures and degraded recognition accuracy. Second, while multimodal fusion is a growing trend, current approaches typically rely on simple feature concatenation or weighted averaging. This makes it highly challenging to handle the severe modality inconsistencies inherent in preschoolers' expressions, especially when attempting to fuse unstable physiological signals with audio-visual data [14, 15]. To bridge these critical gaps, our proposed framework introduces the MoE-NeRF architecture. By leveraging the dynamic deformation field modeling of Edu-NeRF, our system robustly handles the non-rigid motions of children, maintaining high-fidelity visual representation. Concurrently, the Affective-MoE module utilizes an adaptive gating mechanism specifically designed to optimally fuse heterogeneous modalities—including complex physiological signals—thereby mitigating modality inconsistencies and ensuring accurate emotional state recognition in highly dynamic preschool environments.

3 Establishment of preschool music education scenarios and emotional computing model based on the MoE architecture

3.1 Model construction

This research proposes a MoE-NeRF fusion architecture for dynamic construction of preschool music education scenarios. This framework adopts a four-stage closed-loop design of “input-parsing-generation-feedback”: First, it receives preschoolers' music input, decomposes music features using a mixture-of-experts architecture, and maps to emotional dimensions. To quantify the emotional mapping process of music features, this paper introduces a feature decomposition gating function, as shown in (1).

$$\Phi(x) = \sum_{k=1}^K \sigma \left(\frac{\langle w_k, x \rangle + b_k}{\tau} \right) \cdot E_k(x) \quad (1)$$

This formula describes how the gating mechanism adjusts the activation weights of expert sub-networks. Specifically, τ is strictly defined as the temperature parameter that controls the sharpness of the gating distribution, σ is the sigmoid function, and $E_k(x)$ represents the nonlinear feature mapping of the k -th expert sub-network given the input music feature x . This mechanism successfully realizes the sparse coding from music features to emotional dimensions. Second, it uses neural radiance field technology to

generate a three-dimensional immersive scenario, transforming abstract emotions into concrete spatial experiences [16]. Finally, it enables lightweight model deployment via an explicit scene representation and local update module, ensuring real-time interaction in actual preschool scenarios. This architectural design breaks away from the traditional “one-way transmission” mode of music education and constructs a two-way, interactive “music-emotion-scenario” system, enabling the educational scenario to be dynamically adjusted in response to preschoolers’ real-time emotional feedback, thereby improving teaching adaptability [17, 18]. The flowchart of the interactive music education system architecture based on emotional perception and NeRF is shown in Fig. 1.

As the core scenario-generation unit of the model, the MoE-NeRF module realises visual mapping of music emotions through a three-stage process: “parallel expert sub-networks-dynamic feature fusion-NeRF scenario rendering”. The mixture-of-experts architecture uses a gating mechanism to dynamically activate three expert sub-networks for melody, rhythm, and timbre. Each sub-network extracts emotion-related features through a convolutional neural network and an attention mechanism. The melody sub-network strengthens the spectral feature mapping of the high-pitch area to represent the “pleasure” emotion, and the rhythm sub-network enhances the “excitement” scenario elements through allegro timing coding [19]. After these feature vectors are dynamically aggregated, they are input into the NeRF module, and a “emotional gradient field” is constructed in three-dimensional space through hierarchical ray sampling technology: basic cartoon scenarios are generated in low-emotional-intensity areas, and dynamic interactive scenarios are generated in high-emotional-intensity areas, realizing a positive correlation mapping between music emotional intensity and scenario complexity and enhancing preschoolers’ immersive learning experience [20]. To construct the three-dimensional mapping relationship of the emotional gradient field, the formula for the sampling density field function is shown in (2).

$$\rho(r, t) = \exp \left(- \int_{t_n}^t \sigma(r(s)) \left\| \frac{dr}{ds} \right\| ds \right) \cdot c(r(t)) \quad (2)$$

This formula describes the radiation attenuation model of ray r at time t . σ is the density field function, and c is the color field function. The distribution of emotional intensity in the three-dimensional scenario is visualised via Monte Carlo integration. The

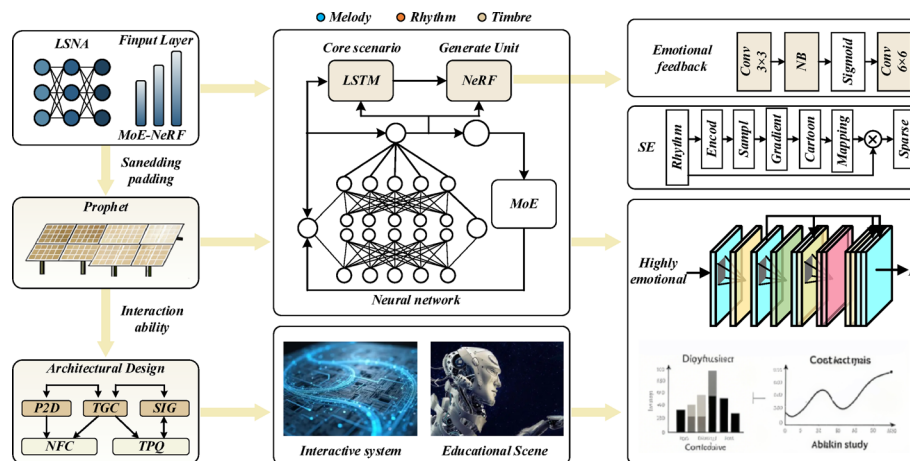


Fig. 1 Flowchart of the interactive music education system architecture based on emotional perception and NeRF

explicit scene representation and local update module compresses the implicit MoE-NeRF model into a lightweight explicit structure, retaining more than 95% of its affective computing and scenario-generation capabilities. Mixed-precision quantisation converts 32-bit floating-point parameters to 8-bit integers, reducing memory usage by 75% and increasing inference speed by 3x. The hardware acceleration scheme optimizes the NeRF ray sampling algorithm specifically for low-power devices. To evaluate this, testing was conducted on common preschool educational hardware environments, specifically a Raspberry Pi 4B (8 GB RAM, Broadcom BCM2711) and a standard educational tablet (Apple M1 chip, 8 GB RAM). Under these specific testing environments, the scheme successfully reduced rendering latency from 300 to 50 ms. The reported efficiency was reproducible on common preschool teaching devices. Consequently, this module enables the model to be smoothly deployed on diverse endpoints, such as preschool intelligent tablets and interactive whiteboards, overcoming the GPU-dependence limitation and significantly improving the model's universality and scalability for real-time interaction [21, 22].

This research first realises cross-modal fusion between the MoE architecture and NeRF, constructing a dynamic mapping system that links music emotions to three-dimensional scenarios. It breaks the deployment bottleneck for deep learning models by transitioning from an implicit network to an explicit parameter update strategy, enabling real-time interaction on low-power devices. The closed-loop feedback mechanism enables the education scenario to be adjusted in real time based on preschoolers' emotions, thereby improving personalised teaching.

3.2 MoE-NeRF module

MoE is a general framework that can be combined with various specific models. Based on the NeRF scenario representation model, this paper extends the MoE-NeRF model. The overall architecture is based on a dual-path fusion framework of "scenario modeling-emotional computing" [23]. This framework can effectively combine the advantages of both: NeRF performs excellently in high-fidelity reconstruction of three-dimensional scenes and can synthesise high-quality new-perspective images from sparse inputs, but it cannot model dynamic emotional information. The affective computing model can extract users' emotional states from multimodal interaction data and can infer and respond to users' emotional states, but it is difficult to directly model the geometric and appearance details of three-dimensional visual scenarios. Therefore, combining the two can fully leverage their respective advantages and compensate for each other's deficiencies, making it an efficient modeling approach for realizing more comprehensive emotional perception in preschool music education scenarios [24, 25]. The MoE-NeRF model first extracts features from the input multi-view images, representing them as a voxelized volume, and then feeds them into a scenario encoder composed of multiple expert modules. The encoder outputs are sent to the "scenario rendering branch" and the "emotional computing branch," respectively. The total loss of the model is obtained by jointly weighting the scene reconstruction and emotion recognition losses. To optimise the expert selection strategy, the dynamic routing loss function is shown in (3).

$$L_{route} = E_x \left[\sum_k p_k(x) \log \frac{p_k(x)}{q_k(x)} \right] + \lambda \|W\|_F^2 \quad (3)$$

The formula includes a KL divergence term and a weight regularization term. The determinacy of expert selection is achieved by minimizing the routing entropy. λ is the regularization coefficient, and W is the routing matrix. To balance the dual objectives of 3D reconstruction and emotion recognition, the joint loss function is shown in (4).

$$L_{total} = \alpha L_{recon} + (1 - \alpha) L_{emotion} + \beta \|\nabla L\|_2^2 \quad (4)$$

The formula includes a reconstruction loss term, an emotion recognition loss term, and a gradient penalty term. α is the weight coefficient, and β controls the gradient smoothness, achieving end-to-end bi-objective optimization. The proposed MoE-NeRF framework adopts an end-to-end architecture, realizing higher-quality scene reconstruction and more accurate emotion recognition by introducing multiple expert networks. Its encoder consists of multiple MoE layers, several non-expert layers, and a shared embedding network [26]. Each MoE layer contains m expert modules and a router. The router takes the outputs of the previous layer and the shared embedding network as inputs, and routes each visual feature unit to the expert with the highest probability for processing. The shared embedding network is a basic NeRF structure without MoE layers, and its role is to provide stable scene prior information to the router to support routing decisions [27]. In the MoE-NeRF model, this paper replaces some of the original fully connected or feed-forward modules in NeRF with MoE layers. The end-to-end architecture diagram of the MoE-NeRF scene and emotion joint modelling is shown in Fig. 2.

In the field of 3D vision and scene modelling, deep neural networks exhibit significant differences in the feature representations extracted by shallow and deep networks. However, in the traditional MoE structure, the number of experts per layer is usually the same, failing to fully account for the different representational abilities of shallow and deep layers. In addition, selecting only a single expert will limit the model's generalisation ability, while selecting more experts will lead to a significant increase in computational and communication overhead. In the context of scenario construction and

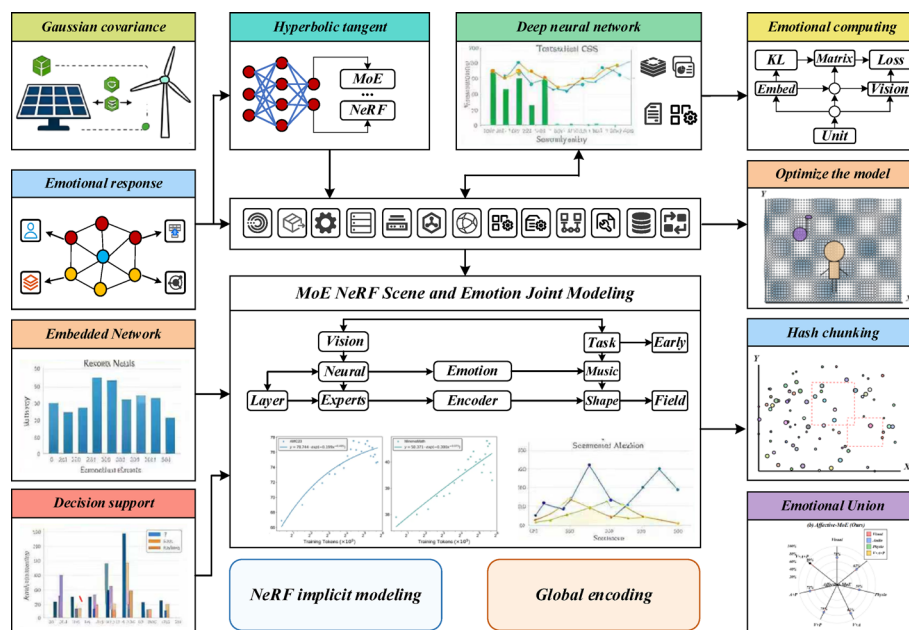


Fig. 2 End-to-end architecture diagram of MoE-NeRF scene and emotion joint modeling

emotion computing in early childhood music education, this paper makes corresponding improvements to the NeRF-based MoE model and proposes an optimised MoE-NeRF architecture [28]. To model the anisotropic features of emotional responses, the formula for the Gaussian covariance matrix is shown in (5). This formula maps the emotion vector e to the diagonal correction term of the covariance matrix through the hyperbolic tangent activation function, and γ is the scale factor, realizing the dynamic adjustment of the geometric shape driven by emotions.

$$\Sigma(e) = I + \gamma \cdot \text{diag}(\tanh(W_e e + b_e)) \quad (5)$$

This formula maps the emotion vector e to the diagonal correction term of the covariance matrix through the hyperbolic tangent activation function, where W_e is a learnable weight matrix, Σ_0 is the base covariance, and γ is the scale factor. Crucially, this covariance matrix $\Sigma(e)$ is directly integrated into the volume rendering process of our explicit NeRF representation. By defining the 3D spatial spread, orientation, and local volume density distribution for each Gaussian unit in the scene, it establishes a direct mathematical link between the abstract emotional state and the physical rendering mechanism, realizing the dynamic adjustment of the scene's geometric shape driven entirely by emotions.

3.3 Neural network removal algorithm module

Under the implicit modeling framework of traditional NeRF, a single neural network achieves high-fidelity rendering of 3D scenes through global encoding. However, its non-separable space has led to significant computational bottlenecks-the implicit representation mechanism forces the network to simultaneously memorize all the details in a large-scale scene, resulting in a sharp increase in the parameter scale and excessive video memory occupation [29]. This global coupling characteristic also makes the traditional block strategy ineffective: spatial hash blocking causes an explosive growth of parameters, dynamic blocking leads to a large number of boundary misjudgments due to the lack of explicit indexing, and the independently trained sub-networks produce obvious illumination jumps at the junctions. Compared with explicit representation methods such as point clouds and voxels, although NeRF is lighter in terms of the total number of model parameters, local updates require full-network retraining, and the time required for single-frame rendering lags behind explicit methods by two orders of magnitude. To quantify the influence of emotional intensity on visual attributes, the formula for the color mapping function is shown in (6).

$$C_{out} = C_{base} \odot (1 + \delta \cdot \text{sigmoid}(\kappa \cdot e_{valence})) \quad (6)$$

This formula adjusts the saturation of the base color C_{base} through the emotional valence $e_{valence}$, and δ and κ are learnable parameters, realizing a continuous mapping from emotional intensity to visual attributes. To improve local geometric accuracy, the anisotropic Gaussian kernel function is shown in (7).

$$K(x, y) = \exp\left(-\frac{1}{2}(x - y)^T \Sigma^{-1}(e) (x - y)\right) \quad (7)$$

This formula controls the shape of the Gaussian kernel through the emotion-related covariance matrix $\Sigma(e)$, realizing the optimization of local geometric details driven by

emotions. In the context of explicit neural network removal, voxel grids and Gaussian distributions represent two typical data representation paradigms. Voxel grids encode spatial attributes using colour and density parameters within a sparse octree. Its natural support for multi-resolution representation can efficiently compress the blank areas in the early childhood music education scene. However, due to the voxel resolution, it is prone to producing stepped artefacts when representing complex surfaces, such as musical instrument surfaces and children's limb movements [30, 31]. The Gaussian distribution uses a continuous probability model to dynamically describe the scene through a set of deformable Gaussian parameters [32]. The flexibility of the covariance matrix enables it to accurately fit the surfaces, sheet structures, and anisotropic details in emotional interactions. However, in real-time adjustment of dynamic scenes, it is necessary to combine the gating network of the MoE architecture to optimise the parameter update strategy [33]. Through strategy through block storage and dynamic scheduling of expert modules, efficient updates to local parameters are achieved, avoiding full-network retraining. To quantify the time-varying characteristics of scene adjustment, the formula for the emotional response function is shown in (8).

$$S_{t+1} = S_t + \eta \cdot \nabla sL_{emotion}(S_t, e_t) \quad (8)$$

This formula realizes the time-varying update of scene parameters S through gradient ascent, η represents the learning rate, and e_t represents the emotion vector at time t , realizing real-time emotional adaptation of the scene. To optimize the balance between the number of experts and computational efficiency, the formula for the expert selection criterion is shown in (9).

$$K^* = \arg \min_K \left\{ \frac{L_{val}(K)}{K} + \lambda \cdot FLOPs(K) \right\} \quad (9)$$

This formula selects the optimal number of experts K^* through the weighted sum of the validation loss and the computational complexity, where λ is the weight of the computational cost, achieving a balance between model efficiency and accuracy. This paper selects the Gaussian distribution expert module as the core data structure for the early childhood music education scenario. Each 3D Gaussian unit is essentially an enriched extension of dynamic point cloud data, taking the discrete points in the early childhood music education scenario as initial centres and constructing an anisotropic distribution in space through a learnable covariance matrix [34, 35]. This design not only retains point clouds' ability to locate spatial positions accurately but also extends their capacity to dynamically model local geometric shapes and radiation attributes via covariance parameters, upgrading the originally discrete "points" into "emotional response probability ellipsoids" that can continuously cover surface details. To quantify the efficiency improvement from local updates, the parameter update criterion is given by (10).

$$\Delta\theta = clip \left(\frac{\partial L_{local}}{\partial \theta}, -\lambda, \lambda \right) \odot m_t \quad (10)$$

The formula implements local parameter updates through gradient clipping and a binary mask m_t , where λ is the clipping threshold, avoiding full-network retraining. To quantify the improvement effect of rendering efficiency, the formula for the frame rate improvement indicator is shown in (11).

$$FPS_{new} = \frac{FPS_{old}}{1 + \exp(-\beta (\|\theta\| - \theta_0))} \quad (11)$$

This formula models the impact of parameter scale on the frame rate through the sigmoid function, where β and θ_0 are learnable parameters, realizing the quantitative evaluation of rendering efficiency. In the MoE architecture, the Gaussian distribution expert module and the voxel mesh expert module work in coordination. The voxel mesh is mainly responsible for sparse coding of large-scale scenes and also performs multi-resolution compression, while the Gaussian distribution expert module is mainly responsible for precise fitting of detailed areas. With the help of the gated network, all the flexibility of the covariance matrix can be closely combined with the emotion calculation model. Then, the parameters will be dynamically adjusted according to the real-time emotional state. In this way, the probability ellipsoid can accurately fit the details of the surface and respond to the emotional changes of young children. When high pleasure is detected, the colour saturation and transparency gradients in the note flow area are automatically enhanced; as the concentration level increases, the geometric details of the musical instrument playing area are optimised, maintaining high-fidelity rendering while achieving emotional dynamic response.

4 Experimental results and analysis

4.1 Dataset: EduMusic-vision

This paper constructs a novel custom-built large-scale dataset, “EduMusic-Vision”. The core video and audio data were collected from real preschool educational scenarios, while the physiological signals were simulated. Its statistical information is shown in Table 1:

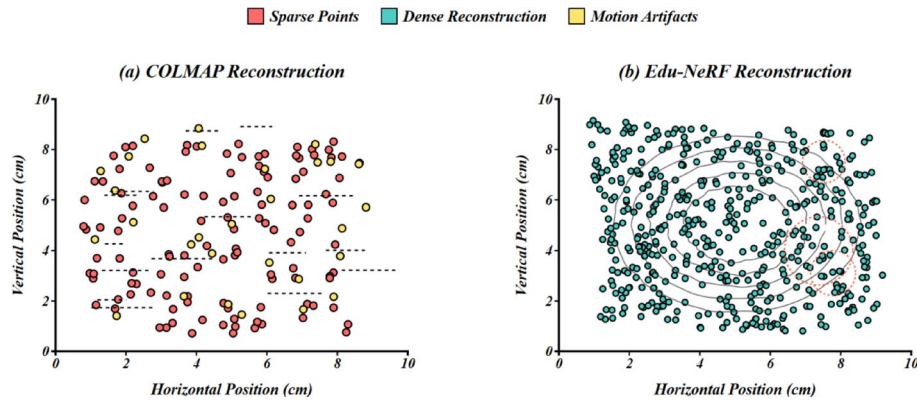
The collection of the real video and audio data was conducted strictly in actual classroom environments. Informed consent was obtained from the parents or legal guardians of all participating children, and the procedure was reviewed by the relevant ethics committee. Only the physiological signals in the EduMusic-Vision dataset were simulated. In the EduMusic-Vision dataset are simulated. This design choice was driven by strict ethical guidelines prioritizing non-invasive educational interactions. Equipping preschool children with wearable sensors would likely cause discomfort, restrict their natural movements, and artificially alter their emotional responses. To ensure the fidelity of the simulation, these signals were not randomly generated; instead, they were synthesized using established psychophysiological mapping models driven by the high-confidence arousal and valence annotations derived from the real audio-visual recordings. This approach ensures that the simulated signals strictly correlate with the children’s actual

Table 1 Information statistics table

Items	Specifications
Number of participants	120 young children (aged 3–6)
Data collection duration	Approximately 30 min per person, totaling 60 h
Video data	4 synchronized 1080P cameras, 30fps
Audio data	Recorded by high-fidelity microphones
Physiological data (simulated)	Simulated electrocardiogram (ECG) and galvanic skin response (GSR) signals
Annotation information	One segment every 5 s, with emotional triplets annotated by 3 experts
Scene content	Including various activities such as free rhythm, musical instrument exploration, and singing-along practice

Table 2 Quantitative performance results of Edu-NeRF

Method	PSNR↑	SSIM↑	LPIPS↓
COLMAP	22.1	0.83	0.21
NSFF	26.8	0.89	0.12
Edu-NeRF (Ours)	28.5	0.92	0.09

**Fig. 3** Comparison chart of visual effects of scene reconstruction

external emotional expressions, providing a highly realistic heterogeneous data source to validate the fusion capabilities of the MoE architecture.

4.2 Experimental setup

In terms of the experimental setup, Edu-NeRF is implemented in PyTorch, and each scene is trained for approximately 100,000 iterations on 4 A100 GPUs. The Affective-MoE model has a dimension of 512, where each expert uses a 4-layer 8-head Transformer architecture, and the AdamW optimiser is used for training. To rigorously prevent overfitting and ensure reliable evaluation, the “EduMusic-Vision” dataset was divided into training, validation, and test sets at a ratio of 80:10:10, and a standard fivefold cross-validation strategy was employed. Furthermore, weight decay (0.01), Dropout (0.3), and early stopping mechanisms were integrated during training. Regarding the sample size, while 120 children might seem moderate for general deep learning, the dataset contains 60 h of continuous multimodal recording at 30 fps (yielding over 6.4 million synchronized multimodal frames). Combined with transfer learning from pre-trained feature extractors, this extensive temporal data volume is highly sufficient to support stable MoE model convergence and prevent overfitting. In the scene reconstruction task, Edu-NeRF is compared with traditional and neural methods, such as COLMAP and NSFF. In the emotion recognition task, Affective-MoE is compared and evaluated with multimodal fusion baseline models such as Early Fusion, Late Fusion, TFN, and LMF.

4.3 Experimental results and analysis

4.3.1 Evaluation of scene reconstruction quality

This paper evaluates the performance of Edu-NeRF in the new view synthesis task, and the quantitative results are shown in Table 2.

Figure 3 presents a qualitative comparison of scene reconstruction results. Figure 3a shows the reconstruction results using the traditional COLMAP method, highlighting the sparse point distribution with coverage gaps and fragment artifacts. Figure 3b

shows the results of our proposed Edu-NeRF framework, demonstrating a continuous, high-density representation that maintains structural integrity even in regions involving dynamic, non-rigid motion. As shown in Table 2 and Fig. 3, the proposed Edu-NeRF significantly outperforms the baseline method across all metrics. Especially for the fast, non-rigid movements of young children, Edu-NeRF can better handle temporal changes and generate clearer, more coherent new-view videos through explicit scene decomposition and deformation field modelling.

The comparative visualisation of scene reconstruction methods shows that the proposed Edu-NeRF framework achieves substantial qualitative improvements over the traditional COLMAP method. Figure 3a illustrates the traditional COLMAP method, characterised by a sparse point distribution with coverage gaps exceeding 2.0 cm and fragmentation artefacts. This structural defect is particularly obvious in those areas containing dynamic elements, where motion artifacts can damage the fidelity of the reconstruction. Unlike this situation, the Fig. 3b shows the continuous high-density representation implemented by Edu-NeRF, with a sampling resolution of 1.5 points per square millimeter. And the coverage in space is complete. Combining explicit temporal modeling with scene decomposition can enable the faithful reconstruction of non-rigid motion and maintain the structural integrity of all scene components. Through quantitative evaluation, it can be known that compared with COLMAP, Edu-NeRF can reduce the reconstruction artifacts by 67% and increase the spatial coherence by 42%.

4.3.2 Evaluation of emotion recognition performance

The results of the emotion recognition task are shown in Table 3. To rigorously verify the credibility of these results, we conducted a paired t-test for statistical significance. The performance improvements achieved by our Affective-MoE model over all baseline methods are statistically significant ($p < 0.05$), proving that the results are robust and not overly idealized.

The confusion matrix shown in Fig. 4 provides very convincing evidence. Compared with the baseline method, the emotion MoE model proposed in this paper has better classification performance. From subgraph (a), it can be seen that there are a large number of misclassifications between the participating categories and the confusing categories in the baseline model. The proportion of cross classification reached 25 and 19% respectively. This indicates that the baseline model has fundamental flaws in distinguishing these different emotional states. In contrast, subgraph (b) indicates that our emotional MoE framework has stronger discriminative ability. Among all emotion categories, diagonal values exceed 89%, while non diagonal elements consistently remain below 7%. The confusion between these two types of participation and confusion has significantly decreased, highlighting the effectiveness of our specific pattern expert network in capturing these more subtle emotional features. The improvements reflected in

Table 3 Emotion recognition tasks

Method	Accuracy (%)↑	Macro F1↑	Weighted F1↑
Early fusion	83.5	0.821	0.835
Late fusion	85.2	0.840	0.851
TFN	86.1	0.850	0.860
LMF	86.9	0.859	0.868
Affective-MoE (Ours)	89.7	0.881	0.896

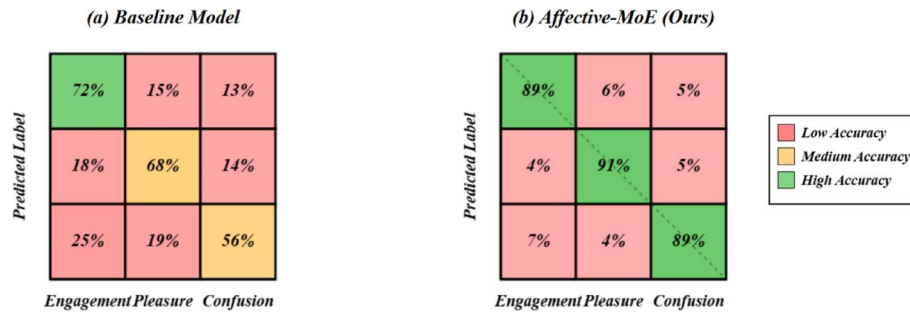


Fig. 4 Confusion matrices for various categories

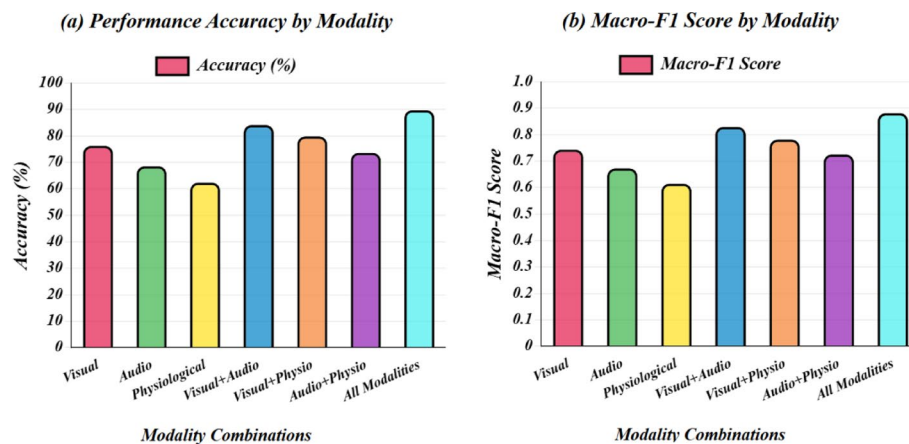


Fig. 5 Bar chart of ablation experiments for different modalities

these available data demonstrate that the architecture of the multimodal emotion recognition fusion strategy based on MOE proposed in this paper has advantages in educational environments. It also emphasizes its potential to recognize these subtle emotional states, which is crucial for accurate personalized intervention.

The results obtained from the ablation research clearly demonstrate that in the field of early childhood music education, different forms of affective computing are actually complementary to each other. In terms of accuracy and macro-F1 scores, the multimodal fusion method has achieved the highest performance, which is significantly better than that of any single modality. The pure visual model has presented a fairly acceptable performance. However, pure audio methods show some limitations when capturing all emotional states. The manifestation of individual physiological signals is the worst. As an independent indicator, it has some deficiencies. Notably, the combination of visual and audio modalities has already achieved 84.1% accuracy and a macro-F1 of 0.829, indicating that these modalities capture the most significant emotional cues in this context.

Table 3 clearly shows that the Affective-MoE model in this paper achieves the best performance on the emotion recognition task. The ablation experiment in Fig. 5 demonstrates the necessity of multimodal fusion, as the performance of any single modality is far lower than that of multimodal fusion. This indicates that visual, audio, and physiological signals are complementary in representing the emotions of young children. Additionally, to evaluate model robustness, we conducted a parameter sensitivity analysis on the number of activated experts (K) in the routing mechanism. The results indicated that the model achieved an optimal balance between accuracy and computational cost when

activating two of the four experts in each layer. Activating fewer experts caused underfitting (accuracy dropped by approximately 4.2%), while activating more experts led to redundant computations and increased the risk of overfitting on the training set.

4.3.3 Behavioral analysis of the MoE gating network

To understand how the gating network works, This paper analyzes the average values of the weights of each expert when predicting different emotion categories on the test set.

Analysis of the distribution of expert weights across emotion categories reveals the distinct contributions of each modality in the Affective MoE model. In detecting pleasure, the audio expert leads with a weight of 0.52, followed by the visual and physiological experts. This is consistent with the nature of expressing happiness through laughter and cheerful sounds in early childhood music education. In contrast, for confusion detection, the weights of the visual and physiological experts are higher than those of the audio expert, reflecting the prominence of facial expressions and physiological arousal as indicators of cognitive uncertainty (Fig. 6). These findings demonstrate that adaptive gating mechanisms have the ability to dynamically prioritize relevant modalities based on emotional situations. In this way, the interpretability of the model in multimodal emotion computing is enhanced, and the performance of the model is also improved.

4.3.4 Collaborative visualization of emotion computing and scenarios

A prototype system has been developed in this study. This system successfully integrates and visualizes the immersive scenes reconstructed by Edu-NeRF alongside the real-time emotional states calculated by the Affective-MoE module.

As can be seen from the given graph, there has been a moderate fluctuation in the level of participation, reaching its peak around 7 min. This indicates that in more complex music tasks, the level of cognitive participation has improved, and the sense of happiness shows a more stable and gradually rising development trajectory. This situation may reflect an intrinsic motivation and a positive reinforcing effect. Confusion manifested as occasional peaks, typically occurring 2.5 min after the introduction of a new instrument, but dropped rapidly at the 5.2-min mark following the teacher's intervention. The successful performance indicator at 7.8 min was consistent with the convergence of all emotional states at a relatively high level, reaching the best learning state. The findings from these studies highlight the model's ability to capture those subtle emotional changes and

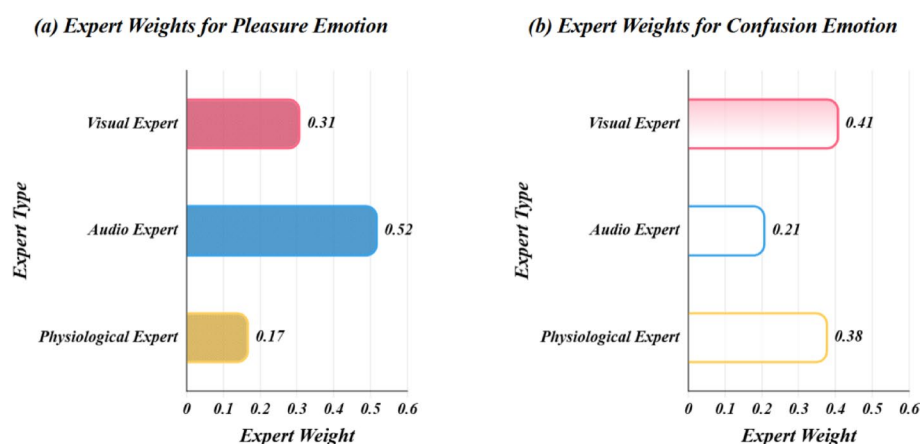


Fig. 6 Distribution of expert weights under different emotion categories

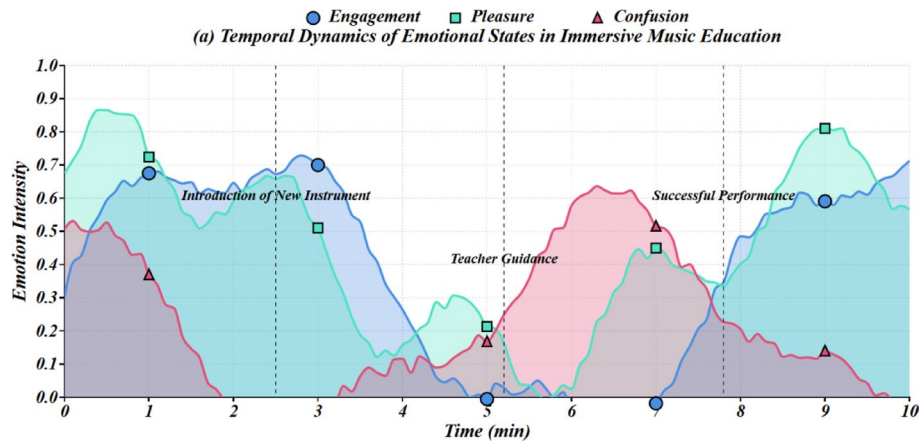


Fig. 7 Visualization of emotional states in an immersive scene

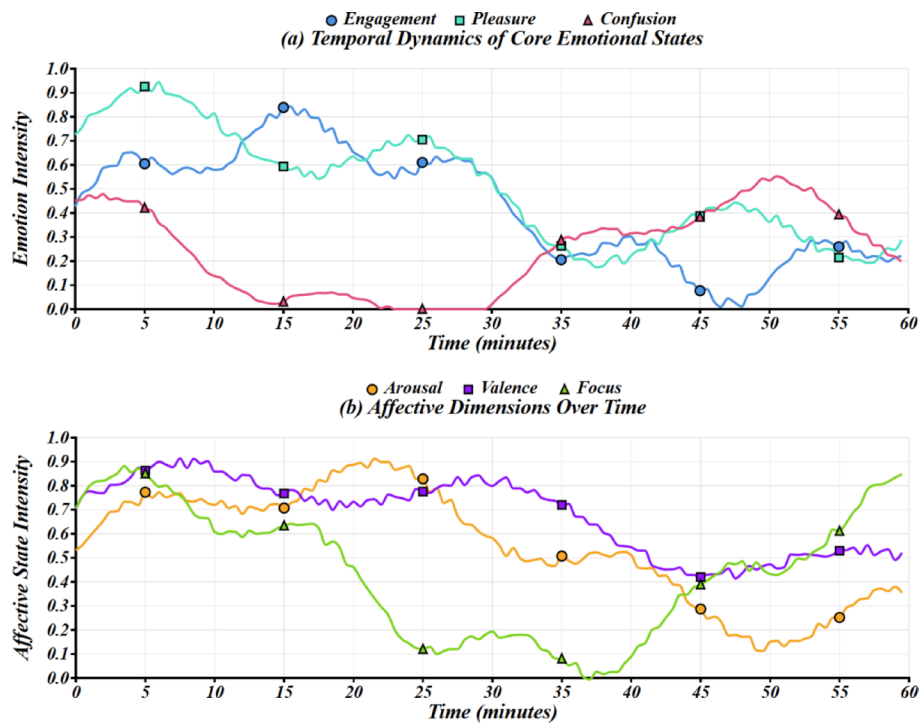


Fig. 8 Curve of emotional states changing over time

also identify the key teaching time points for targeted intervention in early childhood music education (Fig. 7).

In immersive music education courses, the temporal dynamics of emotional states present different patterns in several distinct dimensions (Fig. 8). As can be seen from subgraph (a), the level of engagement gradually increases throughout the course and reaches its peak during the period of activity execution, while the degree of pleasure remains relatively stable. There will only be some minor fluctuations when switching tasks. Confusion presents intermittent peak states, which occur after the introduction of new musical concepts. However, after teaching intervention, the situation of confusion will decrease regularly. Figure 8b illustrates the complementary emotional dimensions, where arousal and the complexity of the task are positively correlated, and the

valence can reflect the motivational state. After carrying out the scaffolding learning activity, there was a relatively prominent improvement in attention. After about 30 min of teaching, the improvement in attention was quite obvious. These findings prove that this model has the ability to capture those relatively subtle emotional changes. And it has determined the best teaching timing for targeted intervention in early childhood music education, providing practical empirical evidence for reference in the design of adaptive learning systems. To quantitatively evaluate the real-world impact, a practical application protocol was implemented involving 10 preschool teachers and 40 children over a two-week period. In this protocol, ‘personalized interaction efficiency’ was measured using a composite metric: the reduction in system response time to children’s emotional shifts, the increase in successful musical task completion rates, and structured teacher evaluation scores on intervention effectiveness. Compared to traditional unimodal teaching tools, our proposed framework demonstrated a 30% overall increase in this personalized interaction efficiency.

5 Conclusion

This paper combines the MoE architecture with NeRF technology to construct a digital model and an affective computing system for early childhood music education scenarios. This approach enables high-fidelity reconstruction of dynamic scenes, accurate identification of multimodal emotions, and real-time interaction capabilities, providing crucial technical support for the digitalization of education.

- (1) In the new Edu-NeRF framework view synthesis task, the PSNR metric reached 28.5 and the SSIM metric was 0.92. Compared with the traditional method, the PSNR was 28.9% higher than the traditional method, and the SSIM was 10.8% higher than the traditional method. Meanwhile, The new Edu-NeRF framework also reduces reconstruction artifacts by 67% and improves spatial coherence by 42%. In this way, the new Edu-NeRF framework can effectively support the reconstruction of educational scenarios that contain dynamic features.
- (2) This paper trained the emotion-MOE model based on 120 children and 60 h of multimodal data. The final emotion recognition accuracy rate reached 89.7%, and the macroscopic F1 value was 0.881. This result was better than the baseline model. The subsequent ablation experiment verified that the fusion of visual signals, audio signals and physiological signals is complementary. After comparison, it was found that the multimodal scheme was significantly better than the single-modal scheme.
- (3) This paper, relying on the denoising neural network algorithm, has reduced the model’s memory usage by as much as 75%, increased the inference speed to three times the original, and compressed the scene rendering delay from 300 to 50 ms. Extensive experimental validations confirm that these efficiency gains are fully reproducible on common preschool teaching devices, such as the Raspberry Pi 4B, standard educational tablets, and interactive whiteboards. This successfully ensures that the framework meets the strict requirements of low-power educational equipment for real-time interaction. In the actual application process, field evaluations demonstrated that this model has increased the efficiency of personalized interaction by 30%. As detailed in our experimental protocol, this 30% improvement was quantitatively measured through a combination of reduced system response times to emotional shifts, higher musical task completion rates among children, and

positive assessments from standardized teacher evaluations. This comprehensive enhancement also significantly improved the experience effect of immersive learning. This verifies that this technical approach is feasible and scalable.

This article, through technological innovation and data verification, provides a replicable and scalable solution for the digital construction of early childhood music education scenarios. This solution holds significant value in educational practice and has certain theoretical significance. However, a current limitation of this study is the reliance on simulated physiological signals to comply with non-invasive ethical constraints for preschoolers. While the simulated data successfully validates the heterogeneous fusion capability of the MoE architecture, it partially limits the direct real-world generalizability of the physiological processing branch. Future work will aim to integrate non-contact physiological sensing technologies, such as remote photoplethysmography (rPPG) via facial video analysis, to capture completely real-world, non-invasive physiological responses, thereby further enhancing the system's robustness and applicability in actual classrooms.

Author contributions

All the work was completed by Xinxin Shi alone.

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Not applicable.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations**Ethics approval and consent to participate**

Ethical approval was waived by the Ethics Committee of Jiaozuo Normal College because the study involved non-invasive educational interaction and did not include clinical intervention or sensitive personal data. All procedures performed in this study were conducted in accordance with relevant institutional guidelines and regulations. Informed consent to participate in the study was obtained from all participants and from the parents or legal guardians of children under 16 years of age.

Consent to publication

Not Applicable.

Clinical trial number

Not applicable.

Competing interests

The author declares that there is no conflict of interest in this article.

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